Lectures in Applied Mathematics
Volume 22, 1985

Some Contributions to the Modelling of Discontinuous Flows

Philip L. Roe

1. Introduction, including the first-order scheme. In 1980 I presented to the Seventh International Conference on Numerical Methods in Fluid Dynamics a paper [1] in which I advanced the rather large claim that "almost any" good numerical technique for solving the linear scalar wave equation ($u_t + au_x = 0$) could be converted, by a fairly simple mechanism, into an equally good numerical technique for solving quasi-linear systems of conservation laws ($\mathbf{u}_t + \mathbf{F}(\mathbf{u})_x = 0$). The particular attraction of that mechanism lay in its ability to generalise the class of asymmetric, or "upwinded" finite-difference schemes, in a simple and natural way. The present meeting seems a good opportunity to review and revise that claim, in the light of $2\frac{1}{2}$ years of additional experience. Some of that experience is my own, and some has been provided by close colleagues. Some has come from contributions made by workers who were able to develop the ideas of [1] in new ways, and some again from people who have quite independently arrived at similar viewpoints and developed related techniques. It must be said at once, however, that all I feel able to do is to place upon these developments a personal and provisional interpretation. The best combinations of these ideas are not yet clear, nor is their final competitive position relative to other classes of methods. Perhaps Bram van Leer's "Ultimate Conservative Differencing Scheme" remains something of a mirage, a dream city hovering in the distant haze. We cannot yet determine how far away it is, even though many of us feel

---

1980 *Mathematics Subject Classification*. Primary 35A40, 35L65, 76L05.

©1985 American Mathematical Society
0075-8485/85 $1.00 + $.25 per page

that its general outline and its directions are becoming steadily more apparent.

I will begin by discussing a fairly general form of the flux difference splitting concept, applicable to one-dimensional conservation laws

$$\frac{\partial}{\partial t}\mathbf{u} + \frac{\partial}{\partial x}\mathbf{F}(\mathbf{u}), \tag{1}$$

where a discrete solution is sought consisting of states $\mathbf{u}_i^n$ equal to the mean value of $\mathbf{u}$, at time level n, inside the ith member of a set of cells. The distinctive feature of flux difference splitting, in my view, is that before advancing to time level $(n + 1)$, we attempt to explain the events taking place at time level n by analysing the "fluctuations" [2], defined as

$$\boldsymbol{\phi}_{i+1/2}^n = \mathbf{F}_{i+1}^n - \mathbf{F}_i^n. \tag{2}$$

For hyperbolic systems, it is natural to "explain" these fluctuations as due to the passage of various waves from one cell to its neighbour (Figure 1). We can view such an explanation as a kind of subgrid model. Evidently the limited data does not make it sensible to seek very elaborate or detailed explanations, nor does it seem very meaningful to ask if one explanation is more accurate than another. What does seem to make sense is to ask for plausible (or, perhaps more accurately, probable) hypotheses which allow us to choose appropriate numerical methods with the least expenditure of computational effort.

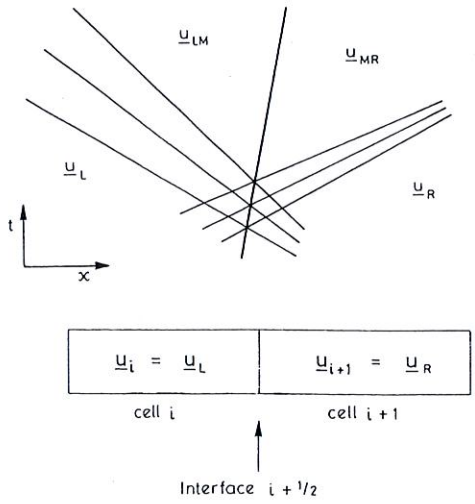


FIGURE 1. Schematic illustration of flux-difference splitting.

My own guess at the present time is that the amount of detail which is worth including in the explanation depends on the complexity of the equations being solved, but that for the unsteady Euler equations a useful explanation comprises the following elements. For each of the three waves $W^{(1)}$, $W^{(2)}$, $W^{(3)}$, we require:

(a) The change of state across $W^{(k)}$.

(b) The average speed of $W^{(k)}$.

(c) The width, or spreading rate, of $W^{(k)}$.

This information can be conveniently arranged into a table;

TABLE I

	$W^{(1)}$	$W^{(2)}$	$W^{(3)}$
Change in F_1 $(= \rho u)$	$\Delta\rho u^{(1)}$	$\Delta\rho u^{(2)}$	$\Delta\rho u^{(3)}$
Change in F_2 $(= p + \rho u^2)$	$(u - a)\Delta\rho u^{(1)}$	$u\Delta\rho u^{(2)}$	$(u + a)\Delta\rho u^{(3)}$
Change in F_3 $(= \rho uH)$	$(H - ua)\Delta\rho u^{(1)}$	$\frac{1}{2}u^2\Delta\rho u^{(2)}$	$(H + ua)\Delta\rho u^{(3)}$
Average wavespeed	$(u - a)$	u	$(u + a)$
Spreading rate	$\delta^{(1)}$	$\delta^{(2)}$	$\delta^{(3)}$

The table has been filled in here with the values which would be obtained analytically, in the case of an infinitesimal fluctuation. In that case we would have

$$\Delta\rho u^{(1)} = \frac{(u - a)}{2a^2}(\Delta p - \rho a \Delta u),$$

$$\Delta\rho u^{(2)} = u\left(\Delta\rho - \frac{1}{a^2}\Delta p\right), \tag{3}$$

$$\Delta\rho u^{(3)} = \frac{(u + a)}{2a^2}(\Delta p + \rho a \Delta u).$$

Here Δ is the difference between state i and $i + 1$, and all other symbols have their usual meaning, except for δ, which will be explained later.

I am going to describe some methods for solving time-marching problems; at each time-step the first action is to draw up a tabular explanation of each fluctuation, somewhat as in Table I. Such tables permit almost automatic extension of numerical methods from the scalar case to the system case. Think of the tables obtained by considering pairs of consecutive cells as being stacked in order like pages in a book. The information digested by a bookworm which selects a particular row (j) and column (k) is the information needed to find the effects of the kth wave on the jth conserved variable. We have split the problem up into

nine nearly independent scalar subproblems. This gives us a great flexibility in designing numerical methods. I believe that in principle a different scalar algorithm could be used on each subproblem. Central differencing for one, upwind differencing for another, an implict method for a third, and so on. Such a combination could certainly be made conservative, and if it proved stable would work after a fashion.

However, it would seem rather pointless to indulge in these variations of technique without good motives. I cannot think of a sensible reason for applying different methods to the subproblems associated with a given wave; one feels that different effects of the same wave should be treated as similarly as possible. On the other hand, there do seem to be reasons for treating each wave independently of the others. The simplest is that algorithms which are stable applied to left-moving waves may not be stable for right-moving waves, and vice-versa. Those algorithms which are stable for both wave directions are suboptimal in many respects compared with those which reflect the appropriate bias. A subtler reason is that different waves may reflect different kinds of nonlinear behaviour. In the case of the Euler equations the sound waves with speed $u \pm a$ are what are called genuinely nonlinear waves, and the associated characteristics may spread to form expansion fans or converge into shock waves. Numerical methods which respond appropriately can be employed once the different types of behaviour have been recognised. The particle paths, whose speed is u, cannot of course collide, and so the associated contact discontinuity has an essentially linear behaviour, which may benefit from being treated by a different algorithm. These refinements were hinted at in [**1**], but have been developed more recently, and will be described below in more detail.

Now although I feel fairly sure that Table I contains about the right amount of information, I am not quite so sure exactly what form that information should take, or how it should be obtained. Taking the second point first, we can solve, exactly if we wish, the Riemann problem for $\mathbf{u}_L = \mathbf{u}_i$, $\mathbf{u}_R = \mathbf{u}_{i+1}$, and find the states LM, MR which appear between the waves in Figure 1. Then at least the first three rows of the table can be filled in, since the change in ρu due to the first wave is $(\rho u)_{LM} - (\rho u)_L$, and so on. There are by now several "approximate Riemann solvers" available. My own technique [**3**] is applicable to ideal compressible flow with an adiabatic equation of state, and its utility has been endorsed in a series of papers by Harten and coworkers [**4–6**]. A rather similar method has been proposed by Pandolfi [**7**]. Extensions to more general equations of state have been given by Lombard et al. [**8, 9**] and by Colella [**10**]. One could also adapt the method of Osher and Solomon [**11**] and of Osher [**12**] to the present purpose. The cited references describe the advantages claimed for each method.

There is also the other issue, of just what information the table should carry. In [13], it is argued that Table II is a better choice.

TABLE II

	$W^{(1)}$	$W^{(2)}$	$W^{(3)}$
Change in u_1 $(= \rho)$	$\Delta\rho^{(1)}$	$\Delta\rho^{(2)}$	$\Delta\rho^{(3)}$
Change in u_2 $(= \rho u)$	$(u - a)\Delta\rho^{(1)}$	$u\Delta\rho^{(2)}$	$(u + a)\Delta\rho^{(3)}$
Change in u_3 $(= e)$	$(H - ua)\Delta\rho^{(1)}$	$\frac{1}{2}u^2\Delta\rho^{(2)}$	$(H + ua)\Delta\rho^{(3)}$
Courant number	$(u - a)\Delta t/\Delta x$	$u\Delta t/\Delta x$	$(u + a)\Delta t/\Delta x$
Spreading rate	$\delta^{(1)}$	$\delta^{(2)}$	$\delta^{(3)}$

This table has also been filled in using the estimates based on small fluctuations; here

$$\begin{aligned}\Delta\rho^{(1)} &= \frac{1}{2a^2}(\Delta p - \rho a \Delta u),\\ \Delta\rho^{(2)} &= \Delta\rho - \frac{1}{a^2}\Delta p,\\ \Delta\rho^{(3)} &= \frac{1}{2a^2}(\Delta p + \rho a \Delta u).\end{aligned} \tag{4}$$

Table I is a rather literal interpretation of the phrase flux difference splitting; Table II will also describe a splitting of the flux difference if we choose to estimate the wave speeds in such a way that

$$\sum_k \nu^{(k)}\Delta\mathbf{u}^{(k)} = \frac{\Delta t}{\Delta x}\Delta\mathbf{F}, \tag{5}$$

where $\nu^{(k)}$ is the Courant number associated with the kth wave. As written, (5) could be taken as a set of three equations to be solved for $\nu^{(1)}$, $\nu^{(2)}$, $\nu^{(3)}$. Alternatively, we may be fortunate enough to find expressions for the wave speeds which make (5), together with (4) and Table II, into an identity. In [13], we find that this does indeed happen (uniquely) for the square root averaging procedure introduced in [3]. This observation makes the following algorithm a conservative differencing scheme.

(a) For every cell we store the conserved variables ρ, ρu, e.

(b) We also store, or possibly compute, u, H, p.

(c) For every pair of consecutive cells (L, R), we compute $\Delta\rho$, Δu, Δp, and the following mean values $\tilde{\rho}$, $\tilde{u}$, $\tilde{H}$, $\tilde{a}$, where

$$\begin{aligned}\tilde{\rho} &= R\rho_L,\\ \tilde{u} &= (Ru_R + U_L)/(R + 1),\\ \tilde{H} &= (RH_R + H_L)/(R + 1),\\ \tilde{a}^2 &= (\gamma - 1)\left(\tilde{H} - \tfrac{1}{2}\tilde{u}^2\right),\end{aligned} \tag{6}$$

in which

$$R^2 = \rho_R/\rho_L. \tag{7}$$

(d) Using these differences and mean values, we fill in Table II, using equations (4).

(e) Let the change in the jth conserved variable due to the kth wave be $\Delta u_j^{(k)}$. Subtract $\nu^{(k)}\Delta u_j^{(k)}$ from the jth conserved variable in the right-hand cell if $\nu^{(k)} > 0$, or the left-hand cell if $\nu^{(k)} < 0$.

Algebraically, this algorithm is completely equivalent to the first-order algorithm described in [**1**,**3**], but in the reformulation the actual flux differences do not explicitly appear. We work with a combination of conserved quantities and primitive variables. This produces slightly faster and neater coding, and gives rise to algebra which is somewhat closer to the more conventional theory of characteristics. In these developments, we have partly followed the ideas of Lombard et al. [**8**, **9**].

Now although the above algorithm, in its original formulation, has been described as Roe's Method, we do not (and never have) recommended it for practical use. Its good features are

g(i) Under many circumstances it produces numerical results [**3**, **4**, **14**] almost identical to those obtained from the classical scheme of Godunov [**15**] but rather more cheaply and rather more systematically. It provides, therefore, an attractive alternative foundation on which to construct more elaborate schemes.

g(ii) It treats slowly-moving shock waves very well indeed. The limiting case of a stationary shock wave is always represented with at most intermediate point [**2**, **16**].

g(iii) Although boundary conditions will not be discussed again in this paper, it is worth noting that if no special action is taken at boundaries, a very satisfactory radiation condition arises by default. Outgoing waves are found simply to disappear without reflection.

The bad features can be listed also.

b(i) First-order accuracy is not sufficient for realistic problems.

b(ii) Rapidly-moving shock waves are badly smeared.

b(iii) Contact discontinuities (unless stationary) diffuse without limit, as time increases, usually like $t^{1/2}$.

b(iv) Nonphysical solutions are obtained if the problem features an expansion fan which crosses a sonic point.

The remainder of this paper will be devoted to overcoming these defects, using only the information already generated in constructing Table II.

The first three faults are closely related. Until a few years ago it was thought to be difficult, if not impossible, to find any type of algorithm which achieved second-order accuracy without introducing unwanted

oscillations near shock waves. We do not attempt to give here a history of this topic, which involves a complex interplay of practical and theoretical considerations. A majority opinion at the present time would probably be that stationary or slowly-moving shocks can be captured adequately by what may perhaps already be described as classical methods; either by adjusting the domain of dependence to suit subsonic or supersonic conditions [**17, 18**], or by devising nonlinear artificial viscosities, of which Jameson's is a recent and ingenious example [**19, 20**]. However, rapidly-moving shocks, across which no characteristic speed changes sign, completely defeat the first of these techniques, and seem so far to have largely eluded the second. More modern techniques all involve some kind of feedback mechanism which adapts the algorithm to the evolving solution. Best known and most widely used are those of van Leer [**21–25**], but see also Harten [**4**], Roe [**26, 27**], Roe and Baines [**28**]. It is this last work which will be followed, and enlarged upon, in §§2 and 3.

In those sections, we shall in fact be relying on the mechanism of [**1**] to extend to systems of equations results which are easily obtainable in the scalar case. This process of extension as tested by numerical experiments has been remarkably trouble-free and straightforward. In §4, we shall turn to the question of entropy-violating solutions. In this respect, the mechanism clearly cannot be complete, since it starts with the linear equations, where questions of physical correctness do not arise. Instead, the mechanism has to be started one stage nearer to its goal, with the nonlinear scalar equations. A modification of the first-order method will be demonstrated which gives physically correct solutions for a class of scalar conservation laws, and the incorporation of this modification, through the mechanism of [**1**], has given satisfactory results for the Euler equations.

2. The "monotone" second-order scheme. I put the word "monotone" in quotation marks to denote a certain deliberate vagueness about the properties I wish to claim. What we really want is a scheme which combines high accuracy with guaranteed exclusion of nonphysical oscillations or entropy-violating solutions. The scheme must apply to systems of equations in many dimensions. We are still very far away from being able to design such algorithms. Concepts such as monotonicity, control of total variation, and monotonicity preservation, all discussed in [**4**], appear to be only interesting preliminary attempts to obtain some mathematical grip on the problem. We follow a pragmatic approach, along the lines indicated by the theory, but often straying beyond the bounds of rigorous proof.

We begin by modifying the first-order algorithm given in §1 in a very simple way so as to achieve second-order accuracy in smooth regions of

the flow. For each pair of cells, i, $(i + 1)$, define

$$\sigma^{(k)}_{1+1/2} = \operatorname{sgn} \nu^{(k)}_{i+1/2}. \tag{8}$$

Step (E) of the algorithm was to subtract, from the value of $\mathbf{u}$ in the cell $(i + \frac{1}{2} + \frac{1}{2}\sigma^{(k)}_{i+1/2})$, the quantity $\nu^{(k)}_{i+1/2}\Delta\mathbf{u}^{(k)}_{i+1/2}$. To develop a descriptive vocabulary, call this cell the target, and the subtracted vector quantity the signal, associated with the kth wave. Let us replace (E) with

(E′) Divide the signal into two unequal fractions, with weights $\frac{1}{2}(1 - \nu)$, $\frac{1}{2}(1 + \nu)$ (for typographical clarity, we have dropped the subscripts), to be subtracted, respectively, from cells i, $i + 1$.

Replacing (E) by (E′) produces an algorithm closely related to that of Lax and Wendroff, to which it reduces if $\mathbf{F} = A\mathbf{u}$ with A a constant matrix. Although the original, physically motivated, target continues to receive the larger share of the signal, the propagation of a "nonphysical" signal counter to the true direction creates typical Lax–Wendroff wiggles in regions of rapidly changing data. So far, therefore, we have failed to profit from the effort which went into explaining the fluctuations. We need to make the scheme a little less symmetrical. To begin with, we merely recast it. We delete (E′) and restore (E), but follow it by

(F) Define the smaller, nonphysical, fraction of the signal due to the kth wave to be

$$\Delta^*\mathbf{u}^{(k)} = \tfrac{1}{2}\big(1 - |\nu^{(k)}|\big)\nu^{(k)}\Delta\mathbf{u}^{(k)}. \tag{9}$$

(G) Subtract this from $\mathbf{u}_{i+1/2-\sigma/2}$; add it to $\mathbf{u}_{i+1/2+\sigma/2}$.

Evidently, (E)–(G) are, together, equivalent to (E′). The key to making the scheme "monotone" is to limit the values of the nonphysical transfers $\Delta^*\mathbf{u}^{(k)}$ in such a way as to rule out overshoots. One way to set about doing this [**28**] is to compare $\Delta^*\mathbf{u}$ in each pair of cells i, $i + 1$, with $\Delta^*_u\mathbf{u}$ in the "upwind" pair of cells i_u, $i_u + 1$, where $i_u = i - \sigma^{(k)}_{i+1/2} = i \pm 1$. Now in smooth regions of the flow, each of these vectors is of order Δx, and they differ by $O(\Delta x^2)$. In the transfer stage (G), $\Delta^*\mathbf{u}^{(k)}$ can be replaced by $\Delta^*_u\mathbf{u}^{(k)}$ and there is no loss of formal accuracy. Nor is there any loss of conservation (since *any* transfers are conservative). If we do make this replacement, we create an algorithm in which each wave system is handled in a manner which is second-order accurate and wholly upwinded, in the sense that if $\sigma^{(k)}$ is of one sign throughout some region, then the ith cell is influenced by the behaviour of the kth wave in the cells i, $i - \sigma^{(k)}$, $i - 2\sigma^{(k)}$. Properties of this scheme, in the constant-coefficient case, are given in, for example, [**17, 29**]. However, this upwinded scheme is not monotone either. A fairly good scheme, but still not monotone in any sense, results from transferring the arithmetic mean of Δ^* and Δ^*_u.

To explain the scheme which we have successfully used on systems of equations, it is probably easiest to revert to the linear scalar case

$$u_t + au_x = 0 \tag{10}$$

for which the Courant number ν is constant, and the nonphysical component of each signal is

$$b_{i+1/2} = \tfrac{1}{2}\nu(1 - |\nu|)(u_{i+1} - u_i). \tag{11}$$

Then at stage (G) we propose to transfer an amount

$$B_{i+1/2} = B\big(b_{i+1/2}, b_{i+1/2-\sigma}\big), \tag{12}$$

where $B(b_1, b_2)$ is some kind of averaging function which can be constructed to meet various desirable criteria. A simple example is the function which returns the value 0 if its arguments are of opposite sign, and otherwise returns the value of the argument with least modulus. Better examples than this can be devised and will form the substance of §3; meanwhile we note that if a B-function has been designed we can, in the spirit of [**1**], apply it to systems like this.

Let $b^{(k)}_{i+1/2}$, $b^{(k)}_{i+1/2-\sigma}$ be suitable norms for $\Delta^*_u\mathbf{u}^{(k)}$ in the cell $i + \frac{1}{2}$ and its upwind neighbour. Then in stage (G), replace the transferred quantity by

$$\Delta^*_u\mathbf{u}^{(k)}_{i+1/2}\frac{B\big(b^{(k)}_{i+1/2}, b^{(k)}_{i+1/2-\sigma}\big)}{b^{(k)}_{i+1/2}}. \tag{13}$$

A good practical choice for the norm appears to be the change in density due to the kth wave. The expression (13) is then easily evaluated using the information contained in Table II. Various B-functions can be supplied to the program as subroutines. It is to the design of these B-functions that we now turn.

3. Designing the B-functions. In this section some desirable properties of B-functions will be developed. We will analyse only the constant-coefficient case (10)–(12), relying on (13) to treat nonlinear systems. We will also make the simplifying assumption that B is a homogeneous function of b_1, b_2; i.e. for all μ,

$$B(\mu b_1, \mu b_2) = \mu B(b_1, b_2), \tag{14}$$

which expresses the desirable property that the algorithm remains invariant if the data is multiplied by a constant. Without real loss of generality we assume $a > 0$. The change of u in the ith cell is due to (i) the signal generated by the cell pair i, $i - 1$, (ii) the nonphysical transfer between $i + 1$ and i, and (iii) the nonphysical transfer between i and

$i-1$. We write these three effects in terms of the fluctuations ϕ (equation (2)), and $k = \Delta t/\Delta x$:

$$u_i^{n+1} - u_i^n = -k\phi_{i-1/2} - \tfrac{1}{2}k(1-\nu)B(\phi_{i+1/2}, \phi_{i-1/2}) + \tfrac{1}{2}k(1-\nu)B(\phi_{i-1/2}, \phi_{i-3/2}). \tag{15}$$

It is apparent that any B-function will give rise to an algorithm which exactly convects constant or linear data. The case of quadratic data will give rise to fluctuations in arithmetic progression, say $\phi - \phi'$, ϕ, $\phi + \phi'$. It is easily shown that the LHS of (15) should then be

$$-k\nu\phi - \tfrac{1}{2}k(1-\nu)\phi', \tag{16}$$

so that quadratic data will be convected exactly if the B-function is such that, for any ϕ, ϕ',

$$B(\phi + \phi', \phi) - B(\phi, \phi - \phi') = \phi'. \tag{17a}$$

This functional equation is satisfied by

$$B(b_1, b_2) = \lambda b_1 + (1-\lambda)b_2, \tag{17b}$$

that is, by any linear averaging function.

For cubic data to be convected exactly, the value which the LHS of (15) should have can be found by cubic interpolation and put into the form

$$-k\phi_{i-1/2} + \tfrac{k}{6}(1-\nu)\big[(1+\nu)\phi_{i-3/2} + (2-\nu)\phi_{i-1/2} - (1+\nu)\phi_{i-1/2} - (2-\nu)\phi_{i+1/2}\big]. \tag{18}$$

This demonstrates that to achieve exact convection of cubics, B must be the particular linear averaging function

$$B(b_1, b_2) = \frac{(2-\nu)}{3}b_1 + \frac{(1+\nu)}{3}b_2. \tag{19}$$

To achieve third-order accuracy in the usual sense, it is sufficient to satisfy (19) within a tolerance of order $(b_1 - b_2)^2$, and to achieve second-order accuracy it is sufficient to satisfy (17b) within a tolerance of order $(b_1 - b_2)$.

However, if B is actually chosen to be an exactly linear average, then the whole scheme is linear, and it is well known (eg. [**15, 22, 27, 4**]) that linear schemes cannot be both monotone and accurate. The use of nonlinear schemes on a linear problem such as (10) does seem a little strange at first, but in fact many other problems are homogeneous rather than linear, for example, finding the maximum value or dominant frequency of a function. In the present context, it is not yet entirely clear what nonlinear behaviour we should be aiming for, but one condition we

can impose is that the increment $(u_i^{n+1} - u_i^n)$ should be bounded between zero and $(u_{i-1}^n - u_i^n)$. This condition [**22, 27**] will certainly avoid overshoots, and if it is taken in conjunction with (15), leads to two constraints upon the B-functions. These are

$$\begin{aligned} \sup[B/b_2] - \inf[B/b_1] &\leqslant 2/\nu, \\ \sup[B/b_1] - \inf[B/b_2] &\leqslant 2/(1-\nu). \end{aligned} \tag{20}$$

These bounds give a clear indication that the ratio between the values of B and its smallest argument should not be permitted to become too large, but the possibility of more delicate constraints needs further investigation. We will outline below some new work which may well be applicable to other methods which make use of nonlinear limiting devices [**4, 25, 30, 31**].

We have already considered the case of fluctuations in arithmetical progression; let us consider the case of geometrical progression. In other words, we have exponentially varying data, with

$$\phi_{i-1/2}/\phi_{i-3/2} = \phi_{i+1/2}/\phi_{i-1/2} = r. \tag{21}$$

In this case, the total increment at the ith grid point can be written, using (14), (15) and (21), as

$$u_i^{n+1} - u_i^n = -k\phi_{i-1/2}\left[1 - \frac{1}{2}(1-\nu)(1-r)\frac{B(r,1)}{r}\right]. \tag{22}$$

If the same exponential data is to be advected exactly by the algorithm, we can easily show that

$$u_i^{n+1} - u_i^n = \frac{r}{a(r-1)}\phi_{i-1/2}(r^{-\nu} - 1). \tag{23}$$

Upon equating the expressions on the RHS of (22), (23) we obtain a rather striking result. If

$$B_\nu(r,1) = \frac{2r}{\nu(1-\nu)}\,\frac{\nu(1-r) + r(1-r^{-\nu})}{(1-r)^2}, \tag{24}$$

then the resulting algorithm will yield the exact solution, after any number of time-steps, for any exponentially varying initial data. Note that (24) implies a more general result. Because of the homogeneous property (14), we have actually defined B as a function of two variables,

$$B(b_1, b_2) = b_2 B(b_1/b_2, 1), \tag{25}$$

but we can most easily illustrate these functions by their intersections with the plane $b_2 = 1$. Note also that (24) defines an algorithm in terms of two quantities of fundamental significance; the Courant number ν, and the parameter r which is often used to describe the smoothness of the data [**22, 27**].

It must be admitted that the algorithm defined by (24) would not be very practical. It would be an expensive computation because of the term $r^{-\nu}$. But (24), despite its rather shapeless appearance, has some intriguing mathematical properties. It is rather easier to analyse (24) after noting that it can be expressed as a hypergeometric function

$$B_\nu(r,1) = r\,{}_2F_1(1,1+\nu,3,1-r), \tag{26a}$$

which converges for $0 < r < 2$. An alternative representation is

$$B_\nu(r,1) = {}_2F_1(1,2-\nu,3,(r-1)/r), \tag{26b}$$

which converges for $\frac{1}{2} < r < \infty$. Note that none of the expressions (24), (26a), (26b) succeed in defining B if r is negative. In fact, most work on limiters up to the present has made some assumption equivalent to setting $B = 0$ if $r < 0$. This causes some loss of accuracy near sharp extrema in the data, which will be discussed later. Meanwhile, we shall adopt this assumption, and note that the monotonicity constraints (20) are thereby simplified to

$$\sup_r B(r,1) \leqslant 2/\nu, \tag{27a}$$

$$\sup_r \tfrac{1}{r} B(r,1) \leqslant 2/(1-\nu). \tag{27b}$$

The function $B_\nu(r,1)$ is plotted against r in Figure 2, for various values of ν in the physically meaningful range $-1 \leqslant \nu \leqslant 2$. (For this range of ν the CFL rule does not prohibit stability.) In all cases B is a monotone increasing function of r, with monotone decreasing first derivative. The following properties of $B_\nu(r,1)$ can be readily established, either from first principles or by appeal to well-known properties of the hypergeometric function [**32**].

(i) The slope at the origin is

$$\frac{\partial}{\partial r} B_\nu(0,1) = \frac{2}{1-\nu},$$

which is the maximum value that it can have if (27b) is to be satisfied.

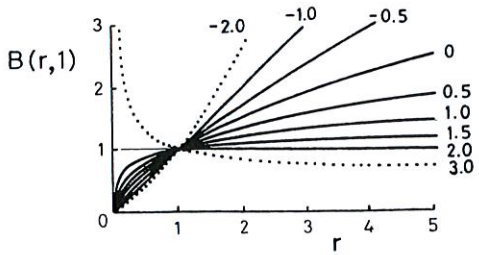


FIGURE 2. The nine special cases of Hyperbee.

(ii) There is a horizontal asymptote whose level is

$$\lim_{r\to\infty} B_\nu(r,1) = 2/\nu,$$

which is the maximum value that it can have if (27a) is to be satisfied.

(iii) $B(1,1) = 1$, and the slope there is

$$\frac{\partial}{\partial r}B_\nu(1,1) = \frac{2-\nu}{3},$$

which is the slope required for third-order accuracy.

Clearly, the B-function defined by requiring the exact advection of exponential data has very distinguished properties. Anyone who is amused by such things may like to observe that the hypergeometric representation passes Kummer's test for the existence of a quadratic transformation [**32**, p. 560] if ν has one of the nine values $\{-2, -1, -\frac{1}{2}, 0, \frac{1}{2}, 1, 1\frac{1}{2}, 2, 3\}$, and then find the corresponding closed form expressions for B_ν. Probably the only useful outcome of that exercise, however, is to note the very simple result

$$B_{1/2}(r,1) = \frac{4r}{(1+r^{1/2})^2}. \tag{28}$$

This result is useful, because $\nu = \frac{1}{2}$ represents a worst case, in the sense that if $\nu = 0$ or 1 the first-order scheme is exact for any data, and the second-order terms vanish because of the factor $\nu(1-\nu)$. The function defined by (28) is tangential, at $r = 1$, to the limiting function introduced by van Leer [**22**] which in our notation is

$$B(r,1) = \frac{r+|r|}{1+r} \tag{29}$$

but (29) does not have any special properties with regard to exponential data. Subsequently van Leer has experimented with many different limiters [**33**, **34**].

Analysis of the special function (24) (hereafter described as Hyperbee, for obvious reasons) began in an attempt to explain some remarkable experimental results, which will be described briefly below, and in more detail in [**35**]. Here we will give the explanation first, and the results later. Suppose we choose some B-function other than Hyperbee, and plot it on the same graph. At any point where the curves intersect, the value of r indicates a special case of exponential data, which would be advected exactly by the algorithm employing that B-function. All legitimate B-functions intersect Hyperbee at $r = 1$. This intersection corresponds to the rather trivial assertion that all our algorithms advect exactly all linear data, so call it the trivial intersection. Most algorithms in use make no

nontrivial intersections, for example the basic choices (see Figure 3)

$$B(r,1) = r \quad \text{(Lax–Wendroff)},$$
$$B(r,1) = 1 \quad \text{(Warming–Beam)},$$

and also the simplest switching algorithm

$$B(r,1) = \min(r,1). \tag{30}$$

The unswitched scheme

$$B(r,1) = \tfrac{1}{2}(1 + r) \quad \text{(Fromm)}$$

generates one nontrivial intersection, unless $\nu = \frac{1}{2}$ when the intersections coincide. The unswitched third-order scheme corresponding to (19) always generates two coincident intersections. A B-function which always generates two widely separated intersections is the following:

$$\begin{aligned} B(r,1) &= \max(r,1) && \text{if } \tfrac{1}{2} < r < 2, \\ B(r,1) &= 2\min(r,1) && \text{for other positive values of } r. \end{aligned} \tag{31}$$

This function, which we have nicknamed Superbee, was devised empirically whilst following up a suggestion by Woodward and Colella [31]

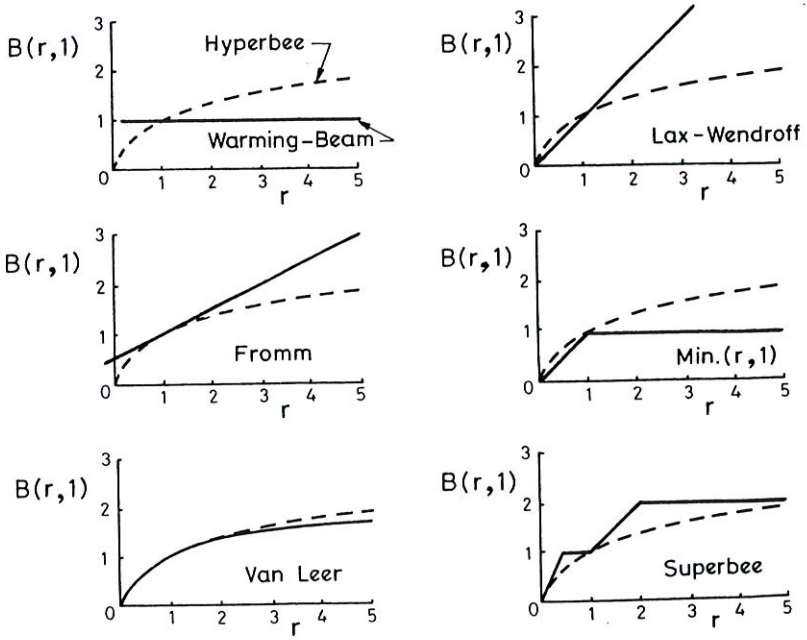


FIGURE 3. Hyperbee compared with six other limiters.

concerning a rather elaborate "discontinuity-sharpening" algorithm. I was trying to implant their idea into my own method, and to give it a simpler expression. In numerical experiments involving step data for the linear equation (10), it was found that the use of Superbee caused stable profiles to form which propagated essentially unchanged for many thousands of time steps (see Figure 4). When the numerical values were investigated, it was found that the head and tail of the profile corresponded rather closely to the values of r found at the two nontrivial intersections between Hyperbee and Superbee. The inflexion region which joins them is difficult to analyse theoretically but behaves very simply in practice. It just stays smooth and does not grow.

Some properties of Superbee have been investigated by Sweby [**36**], who also gives interesting results for test problems involving linear and nonlinear advection, and also for the Sod shock-tube problem. Along the lines of the discussion following Table I, he argues that the best combination of algorithms is to deploy Superbee on the second wave (or particle path) and (29) or (30) on the nonlinear waves. He also discusses the relationship between these limiters and the ones proposed by Harten [**4**] and by Chakravarthy and Osher [**38**].

Superbee has been applied to another one-dimensional test problem by the present author with the assistance of Paul Glaister (University of Reading). This is the blast-wave interaction problem proposed by Woodward and Colella [**39**]. A late stage in the evolution of this problem is shown in Figure 5, where density and velocity are both plotted against distance along a one-dimensional tube. Shock discontinuities are present in both plots, but contact discontinuities are only present in the density plots. Results using Superbee to compute all three wave fields are compared with results using (30) in the same way. A marked improvement of the contacts is noticeable when Superbee is used.

Lastly, we report that Superbee has been successfully incorporated into the two-dimensional aerofoil code [**40**] written at the Royal Aircraft

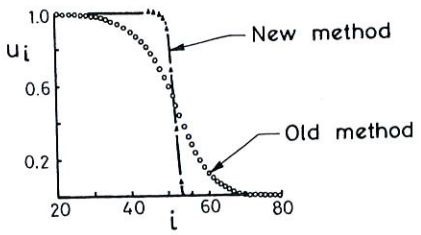
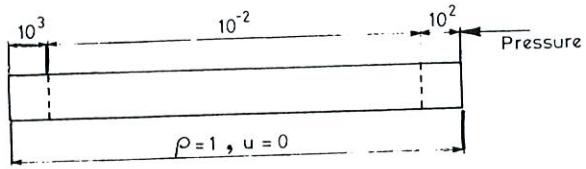
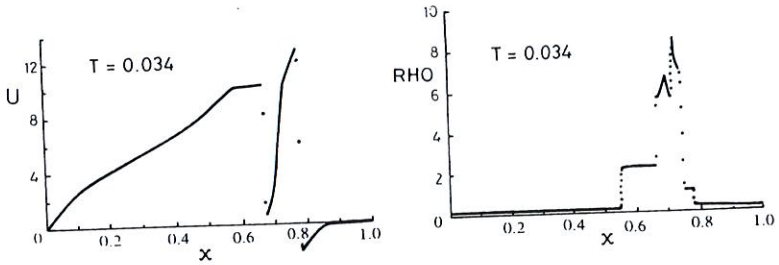


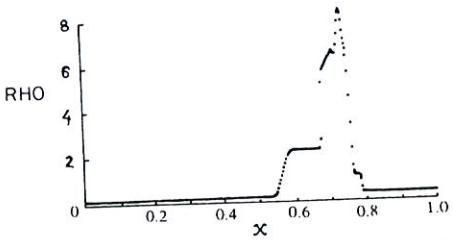
FIGURE 4. Linear advection of a step-function: 10,000 time steps at $\nu = 0.1$.



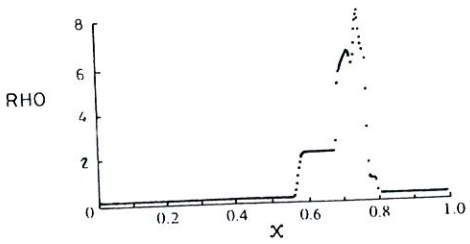
Initial conditions with discontinuities at $x/L = 0.1, 0.9$.



(a) Solution by present method with superbee: 1200 cells.



(b) Solution by present method with equation (30): 400 cells.



(c) Solution by present method with superbee: 400 cells.

FIGURE 5. The computed interaction between two blast waves.

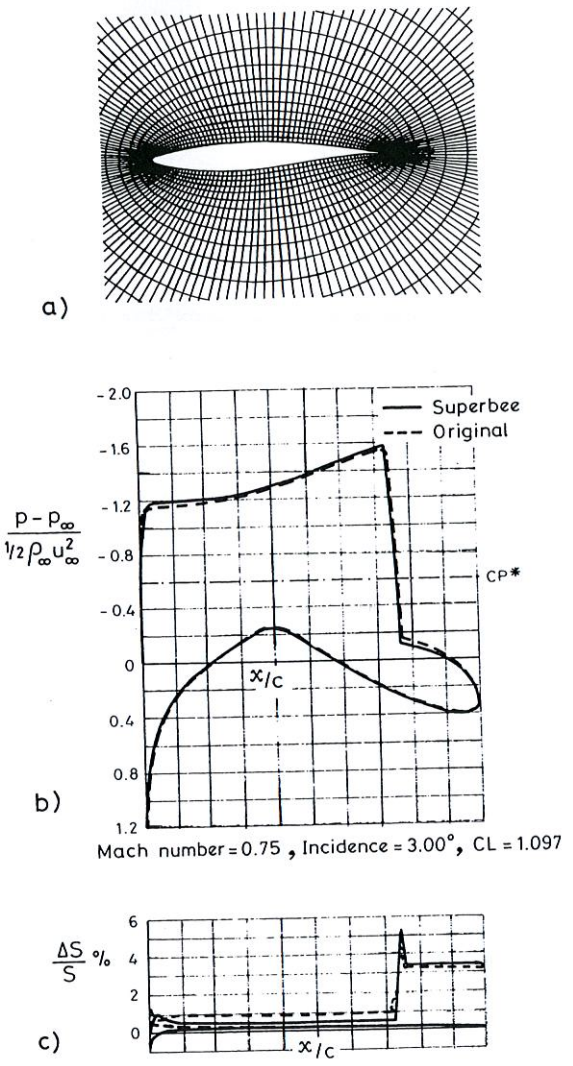


FIGURE 6. Transonic flow past a two-dimensional aerofoil (RAE 2822). (a) Part of computing grid (b) Surface pressure (c) Surface entropy

Establishment by Dr. C. C. Lytton (formerly Dr. C. C. L. Sells). He finds [41] that changing from a limiting procedure based on (30) to one based on Superbee has significantly altered the results obtained from his code. A typical experience is that gained on computations of the flow past on RAE 2822 aerofoil at a freestream Mach number of 0.75 and an incidence of 3°. This is a standard AGARD test case [42]. The aerofoil and part of the 160 × 24 0-grid mesh are shown in Figure 6(a). The following changes in the converged steady-state results were observed.

(i) The lift coefficient increased fractionally, from 1.0948 to 1.0974.

(ii) Originally there was a substantial discrepancy between two different ways of estimating the drag coefficients. An estimate based on integrating surface pressures gave 0.0479, and an estimate based on evaluating energy degredation through the shock gave 0.0385, a discrepancy of some 20%. The revised estimates are 0.0461 and 0.0452, with a discrepancy of about 2%.

(iii) The pressure distribution over the aerofoil surface has changed visibly (Figure 6(b)). Whether the overall distribution is an improvement or not is of course nearly impossible to determine, but the shock wave is slightly stronger and a little crisper.

(iv) The surface entropy distribution is definitely improved (Figure 6(c)). The quantity plotted is

$$\frac{p}{\rho^\gamma}\frac{\rho_\infty^\gamma}{p_\infty} - 1$$

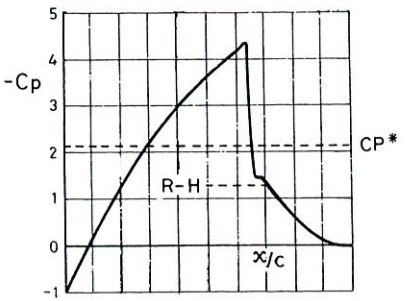
which should of course be zero ahead of the shock wave. This particular error has been reduced by a factor rather better than 2.0.

Putting all this evidence together, one is left with the definite impression that some slightly dissipative effects have been removed.

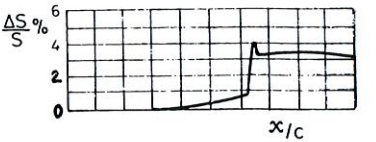
Another test has involved computing the supercritical flow past a semicircular obstacle, a problem recently tackled by several authors [43]. The problem is attracting attention because of the observation that many Euler codes spontaneously generate solutions containing regions of separated or recirculating flow. It is widely conjectured that these solutions may be good approximations to the limiting solutions of the Navier–Stokes equations at large Reynolds number. However, if closed streamlines are present, this limit may not be unique [44]. Therefore, if the results of different codes are compared, it is difficult to find grounds for preferring one to another. Here, I merely record as a matter of possible interest, that the code which incorporated (30) used to fail if asked to compute the flow at any Mach number much greater than 0.40. It is at about this Mach number that supersonic flow appears near the equator, and separated flow almost simultaneously occurs near the rear stagnation point. Using Superbee, supercritical flows with closed separations have been found at $M_\infty = 0.50$ and 0.60. These are the only

supercritical cases which have been attempted so far. Some results are shown in Figure 7. Note that the positions of the separation points have not been fixed in any way. The recirculating regions develop spontaneously in the course of a normal run.

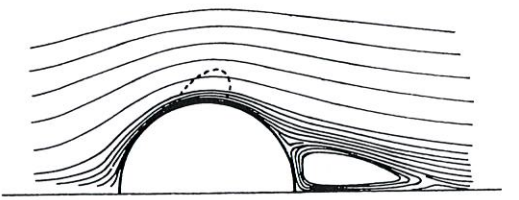
4. On satisfying entropy. It is very well known that some finite-difference schemes admit stable steady solutions which do not satisfy any of the tests by which the physically correct members are selected from the



(a) Surface pressure.

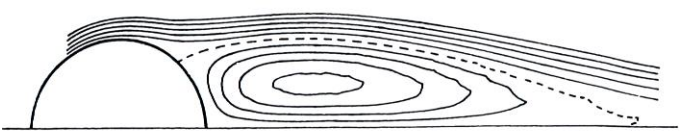


(b) Surface entropy.



(c) Streamlines and sonic line.

7.1. Freestream Mach number = 0.50.



7.2. Freestream Mach number = 0.60.

FIGURE 7. Computations of transonic flow past a circular cylinder.

set of weak solutions. See, for example, [2, 4, 11, 16, 45, 46]. It is traditional to illustrate this point by considering the scalar case (the theory for systems is still very undeveloped)

$$u_t + F_x = 0, \tag{32}$$

and then specialising to Burger's equation, $F = \frac{1}{2}u^2$, and initial data $u(x, 0) = \text{sgn}(x)$. This problem admits the physically reasonable solution

$$u(x, t) = \begin{cases} \text{sgn}(x) & \text{if } |x|/t > 1.0, \\ x/t & \text{if } |x|/t < 1.0, \end{cases} \tag{33}$$

and also the solution

$$u(x, t) = \text{sgn}(x) \tag{34}$$

everywhere, which is physically unreasonable because characteristics emerge out of the discontinuity along $x = 0$. Solutions like (34) are said to violate an entropy condition. Now we consider the discrete problem on some rectangular grid as in previous sections, and propose initial data $u_i^0 = -1$ if $i < 0$, $u_i^0 = +1$ if $i \geqslant 0$. Many finite-difference schemes, including all the schemes described so far, will accept this data as being already in equilibrium (because all the fluctuations are zero) and will therefore propagate it without change to all subsequent time levels.

My excuse for mentioning this very hackneyed example is to invite the reader to consider a slight twist to the story. Suppose the differential equation had actually involved the flux function $F = 1 - \frac{1}{2}u^2$? The numerical data would remain unaltered, whether considered as a set of u-values or as a set of F-values, but it would now comprise a valid, realistic, steady solution, representing the exact continuum solution with maximal resolution. So, what kind of finite-difference scheme will correctly distinguish between the two cases?

It seems to me that there are essentially two possibilities. One is that the scheme contains some dissipative element which widens the transition zone, and introduces values of u not present in the initial data. The computer now has more information about the function $F(u)$. However, there are still infinitely many functions $F(u)$ which coincide with the data at the discrete samples, and so the computer still cannot be sure that we have not chosen an $F(u)$ requiring an expansion fan. It has no option but to widen the zone still further, introducing new values of u, and acquiring more information about $F(u)$. A typical example of an algorithm which can be proved to converge, in this sense, to an entropy-satisfying limit, as the grid is made finer is that of Lax and Friedrichs [47]. However, no algorithm which operates like this can admit stationary shocks with steady profiles of finite width, and the Lax–Friedrichs scheme would today be considered far too dissipative for practical use.

The other possibility is that the computer is supplied directly with more complete information about the function $F(u)$. The prototype for such an algorithm is Godunov's scheme [**15, 46**] where the information derives from exactly solving the Riemann problem for all pairs of adjacent states. It is not possible to find this exact solution unless $F(u)$ is known for all values of u between the given values. It might be said that the computer is, in effect, supplied with a device which allows it to look inside the intervals without opening them up. Actually, for the scalar case, which is what we are discussing, this strategy is not as hard as it may seem. Osher [**48**] has shown that $F(u)$ can be sufficiently characterised by knowing all its extreme values; the only reason why an algorithm of the first kind must continually open up the intervals is the danger that a new extreme value may await discovery.

Osher's result is expressed in terms of interface flux functions $F_{i+1/2}$, such that

$$u_i^{n+1} - u_i^n = -\frac{\Delta t}{\Delta x}\left(F_{i+1/2}^n - F_{i-1/2}^n\right), \tag{35}$$

and states that convergence to an entropy solution is guaranteed by the following condition:

$$\begin{aligned} &\text{if } u_{i+1}^n > u_i^n, \text{ then } F_{i+1/2} \leqslant \inf F_{i+1/2}, \\ &\text{if } u_{i+1}^n < u_i^n, \text{ then } F_{i+1/2} \geqslant \sup F_{i+1/2}, \end{aligned} \tag{36}$$

where $\sup F$, $\inf F$ are, respectively, the greatest and least values assumed by F in the range $\{u_i, u_{i+1}\}$. Osher calls such schemes E-schemes.

It is easy to translate this result into the present fluctuation-signal approach. Let each fluctuation $\phi_{i+1/2}$ be divided, with an obvious notation, into a left moving part and a right moving part; thus, dropping the n suffix

$$\phi_{i+1/2} = \overleftarrow{\phi}_{i+1/2} + \vec{\phi}_{i+1/2}, \tag{37}$$

where

$$\overleftarrow{\phi}_{i+1/2} = F_{i+1/2} - F_i, \tag{38a}$$

and

$$\vec{\phi}_{i+1/2} = F_{i+1} - F_{i+1/2}; \tag{38b}$$

then (36) implies that if $u_{i+1} > u_i$,

$$\overleftarrow{\phi}_{i+1/2} \leqslant \inf F_{i+1/2} - F_i, \qquad \vec{\phi}_{i+1/2} \geqq F_{i+1} - \inf F_{i+1/2}, \tag{39a}$$

whereas if $u_{i+1} < u_i$,

$$\overleftarrow{\phi}_{i+1/2} \geqq \sup F_{i+1/2} - F_i, \qquad \vec{\phi}_{i+1/2} \leqq F_{i+1} - \sup F_{i+1/2}. \tag{39b}$$

It is easy to find geometric interpretations for these formulae, but there are several reasons why it may not be profitable to pursue these results too closely or too rigorously. An objection of principle is that a theory which applies only to scalar problems has at present to be extended by guesswork anyway. An objection of practice is that some of the schemes which can be demonstrated to converge to physically correct solutions actually do so in a rather disappointing way. Particularly it should be said that Godunov's method, sometimes thought of as the ideal 'physically based' scheme, actually gives expansion fans which have almost discontinuous behaviour at sonic points [**5, 39, 50**]. Also we cite the recent experience of Harten [**4**], who states that 'with reluctance' he has abandoned an earlier, rigorous approach to entropy enforcement [**5, 46, 49**] in favour of an empirical device which gives better numerical results for the Euler equations. We present below an account of a "spreading device" which has also proved successful in practice, but is motivated by physical arguments and contains no arbitrary parameters.

We begin by seeking estimates for the spreading rates associated with each wave in each fluctuation, i.e. the quantities $\delta^{(k)}$ in Tables I and II. For definiteness, consider $\delta^{(1)}$. Many estimates suggest themselves, and I suspect that it does not, in most cases, make much difference which we choose. For example,

$$\delta^{(1)}_{i+1/2} = \frac{1}{2}\left[(u-a)_{i+3/2} - (u-a)_{i-1/2}\right], \tag{40}$$

making use of wave speeds worked out for neighbouring interactions, or

$$\delta^{(1)}_{i+1/2} = (u-a)_{i+1} - (u-a)_i. \tag{41}$$

There are mild objections to both estimates. Equation (40) is less "local" than it could be, and (41) puts us to the trouble of computing wave speeds in each cell, which is expensive information for which we have no other use. A third estimate comes from considering the expansion fan, if it exists, to be a simple wave, across which the Riemann invariant does not change

$$a + \frac{\gamma - 1}{2}u = \text{constant}.$$

From this it follows at once that

$$\delta^{(1)} = \Delta(u-a)^{(1)} = \frac{\gamma+1}{2}\Delta u^{(1)}, \tag{42a}$$

and similarly

$$\delta^{(3)} = \Delta(u+a)^{(3)} = \frac{\gamma+1}{2}\Delta u^{(3)}, \tag{42b}$$

whilst, on physical grounds,

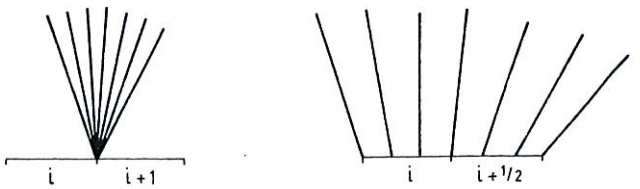
$$\delta^{(2)} = 0. \tag{42c}$$

These estimates use only local information which has already been generated. If (42a) or (42b) should turn out to be negative, then we know that we are dealing with a compression, and we set that spreading rate to zero. The part of the algorithm which deals with that wave will then be as described above, as of course will the part dealing with the particle path. A neat way to find the velocity jumps [**13**] is

$$\Delta u^{(1)} = \frac{1}{2}\left[\Delta u - \frac{\Delta p}{\tilde{\rho}\tilde{a}}\right], \tag{43a}$$

$$\Delta u^{(3)} = \frac{1}{2}\left[\Delta u + \frac{\Delta p}{\tilde{\rho}\tilde{a}}\right]. \tag{43b}$$

Now we make the (slightly arbitrary) decision that the standard form of the algorithm will also be applied to any expansion wave which runs wholly in one direction. (There is some justification for this; all the numerical results shown so far have completely neglected the distinction between compression and expansion waves.) Thus, no special action will be taken unless the spreading rate is positive and greater than twice the modulus of the mean wave speed. In that case only, we shall modify the *first-order* version of the algorithm, causing the fluctuation to send a signal to both targets, as in (37). If we assume that the characteristic field is centered at the interface between the cells (Figure 8(a)) it can be shown that we get an effect proportional to the square of the spreading rate. This is similar to the effect considered, and rejected, by Harten. However, if we assume that the characteristic field varies linearly across the cells (Figure 8(b)), then we find an effect directly proportional to the spreading rate. Not only is this a larger effect but also probably a more



(a) Centred. (b) Distributed.

FIGURE 8. The representations of a rarefaction.

realistic effect, it being in the nature of expansion fans to spread out and assume a fairly linear distribution. In fact, we obtain

$$\overleftarrow{\phi} = \frac{1}{2}\left(1 - \frac{1}{2}\frac{\delta}{\lambda}\right)\phi, \tag{44a}$$

$$\vec{\phi} = \frac{1}{2}\left(1 + \frac{1}{2}\frac{\delta}{\lambda}\right)\phi, \tag{44b}$$

where δ is the spreading rate, and λ is the mean velocity. The same concepts can be expressed in terms of an interface flux function,

$$\begin{aligned} F_{i+1/2} &= \frac{1}{2}\left(1 + \frac{1}{2}\frac{\delta}{\lambda}\right)F_i + \frac{1}{2}\left(1 - \frac{1}{2}\frac{\delta}{\lambda}\right)F_{i+1} \\ &= \frac{1}{2}(F_i + F_{i+1}) - \frac{\delta}{4\lambda}(F_{i+1} - F_i) \\ &= \frac{1}{2}(F_i + F_{i+1}) - \frac{\delta}{4}(u_{i+1} - u_i) \\ &= \frac{1}{2}(F_i + F_{i+1}) - \frac{1}{4}(F'_{i+1} - F'_i)(u_{i+1} - u_i). \end{aligned} \tag{45}$$

Equations (44a), (44b) and (45) apply very simply to Burger's equation. We get

$$\overleftarrow{\phi} = \tfrac{1}{2}u_i(u_{i+1} - u_i), \tag{46a}$$

$$\vec{\phi} = \tfrac{1}{2}u_{i+1}(u_{i+1} - u_i), \tag{46b}$$

$$F_{i+1/2} = \tfrac{1}{2}u_i u_{i+1}. \tag{47}$$

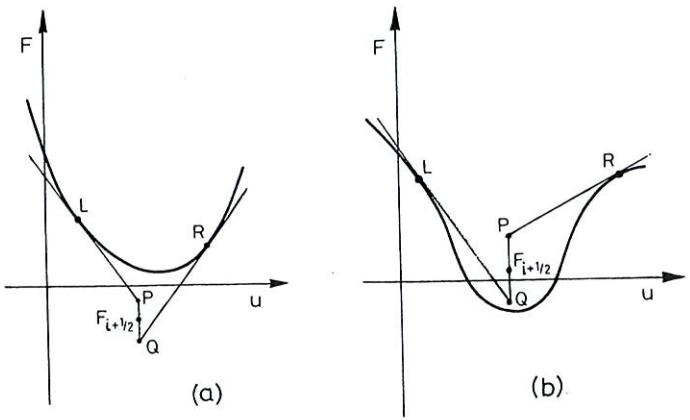
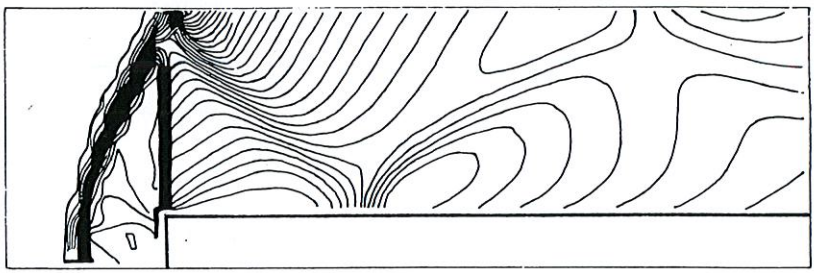
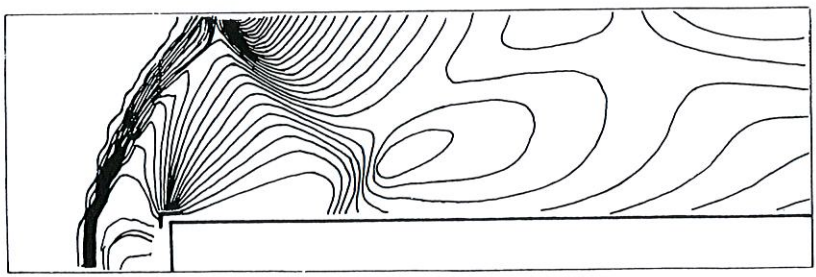


FIGURE 9. Examples of function for which equation (40) is (a) or is not (b) an E-scheme.

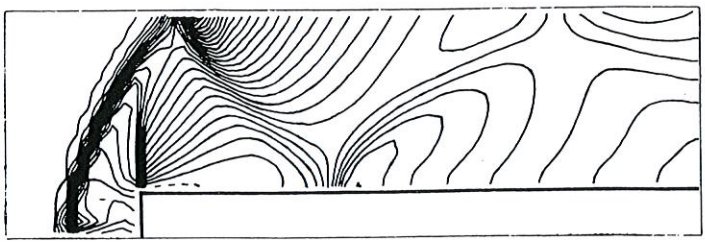
Equation (47) is to be inserted into the usual upwind scheme only if $u_i < 0 < u_{i+1}$, in which case $F_{i+1/2} < 0$, making the scheme an E-scheme, in Osher's sense [**48**]. Very satisfactory numerical tests were reported by van Leer in [**14**]. For more general equations, however, (45) may not be an E-scheme, as may be seen by a geometric representation (Figure 9(a)). LP, RQ are tangents to $F(u)$ at u_L, u_R; P, Q are at $\frac{1}{2}(u_L + u_R)$; $F_{i+1/2}$ is the midpoint of PQ, to be used if $F_i' < 0 < F_{i+1}'$. This makes (45) an



(a) Present method: first order, no spreading of expansion waves.



(b) Present method: first order, expansion waves spread.



(c) Godunov's method [**39**].

FIGURE 10. Three versions of the unsteady flow in a stepped channel; grid size 20 × 60.

E-scheme if $F(u)$ is any concave or convex function, but clearly it is possible to devise $F(u)$ such that (45) is not an E-scheme (Figure 9(b)).

Generalising the scheme to systems can be done very easily starting with (44a) and (44b) and applying the formulae to each column of Table II. That something along these lines may be necessary in practical cases can be seen by modifying the popular Sod test problem [**37**] by superposing on the initial conditions a uniform translation velocity of half a unit. Computations by Harten [**4**] and Sweby [**50**] show an unrealistic discontinuity in the middle of the expansion. For a two-dimensional example we exhibit Figure 10, due to Glaister [**51**], showing computations of another test problem proposed by Woodward and Colella [**39**]. This is the flow brought about by impulsive introduction of a step into a wind tunnel containing uniform flow at $M = 3.0$. The results are a snapshot, after the uniform flow has had time to travel four-step heights. Figure 10(a) is a simple first-order time-split computation, with no special action taken at sonic points, whereas Figure 10(b) incorporates a scheme very similar to the above and due to Sweby [**50**]. In Figure 10(a) the subsonic region is terminated by a one-dimensional expansion shock. Remarkably, when we look at Figure 10(c), taken from [**39**], the calculations made according to Godunov's method are nearly as bad as those in Figure 10(a).

Sweby has extended his scheme to second-order accuracy [**50**], but it seems quite possible that a first-order scheme may be good enough. The heuristic argument is that if we write the scalar equation as

$$u_t + a(u)u_x = 0, \tag{48}$$

then, in the cells adjacent to a smooth sonic point, $a(u)$ is of order Δx and so u_t is small anyway. If the sonic point is not in a region of smooth flow, then of course the formal accuracy of the scheme is irrelevant. In either case u_t has been estimated as accurately as we can justifiably demand. The tie-in with the programming logic of §2 is to set $\Delta^*\mathbf{u}^{(k)}$, as defined by (9), equal to zero, if the kth wave is an expansion containing a sonic point. Incidentally, the same heuristic argument suggests that we could afford to lose one order of accuracy near a smooth extremum (see the comment following (26b)).

Whilst on the general subject of guaranteeing the correct physical solution, I would like to draw attention to another feature of Figure 10. This is the reflection of the shock wave from the upper surface of the step, following its first reflection from the tunnel roof. In many of the computations reported by Woodward and Colella [**39**], this reflection takes place in a regular way. In Glaister's calculations, the reflection is regular when it first occurs, but develops with the passage of time into something more resembling the Mach reflection [**52**] which appears on

the roof in both sets of calculations. This second Mach reflection is also seen in the preliminary results of Harten [4]. Which type of reflection is correct?

Let us note that something like Mach reflection is what is observed in experiments when the flow is viewed in sufficiently small detail [53], because there is always a thin boundary layer of low-energy gas. Now consider how the solid-wall boundary condition will be implemented in this particular problem. Since all boundaries are plane, it would seem sufficient to use fictitious image cells just outside the real flow, where pressure, density and parallel velocity are all set equal to the corresponding quantities just inside the real flow, but the normal velocity has its sign reversed. Indeed the only place where we might anticipate a problem is at the upper corner of the step, where the image cell has two different jobs to do.

I do not want to discuss the 'correct' way out of this local difficulty. Experience shows that it is easy to go wrong, but theory suggests that it will not be easy to tell if we have got it right. The exact inviscid solution around this corner may well be quite complicated, with a closed separation downstream of which the flow passes through an embedded shock wave (Figure 11). If this does happen, the final steady solution is probably not unique. Even if it does not happen, the body will be sheathed in an entropy layer, or entropy wake [54], of low density gas, which has passed through the strongest part of the bow shock wave. This layer might be expected to interact with the incident shock wave much as

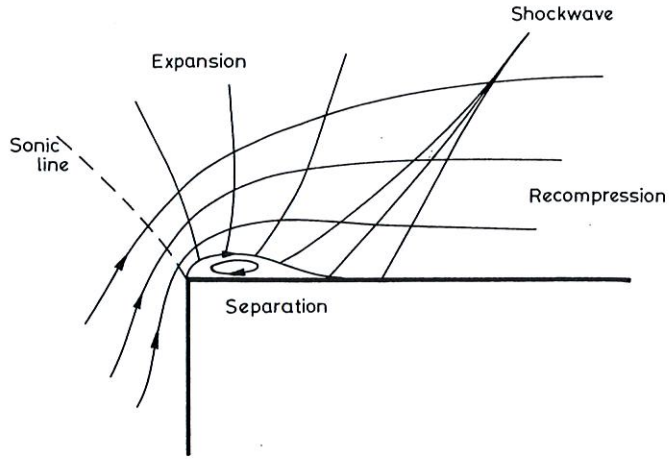


FIGURE 11. Possible high Reynolds number flow pattern around corner of step.

a boundary layer does. The point I wish to make is that even for relatively simple flows, it can be very hard to decide whether plausible looking results are actually reliable. These problems do not seem to be resolved by appeal to an entropy principle, or any other general principle that I know of.

5. Final remarks. At the time of writing [1], I hoped that a mathematical structure was revealing itself which would be sufficiently powerful to provide a complete link between studies of one-dimensional numerical advection, and completely arbitrary one-dimensional quasi-linear conservation laws. Today I am not quite so optimistic. The main theoretical obstacle to handling all conservation laws by one single method is the need to account for all possible modes of nonlinear behaviour. If these modes are complicated, then the scope for using approximate Riemann solutions is restricted. For if the approximate solution completely ignores the internal wave structure, as in [3], then some additional information must be added on. On the other hand, if the approximate solution aims also at describing the internal structure, as in [11], such a solution is unlikely to emerge in a computationally inexpensive form, except in the simplest cases. For some time to come, the compromise between economy and reliability probably has to be resolved individually for each set of equations we might wish to solve.

Where these equations have been well studied, however, and it is known that the modes of nonlinear behaviour are rather simple, then matters seem much clearer. The most valuable concept seems to be that of splitting the interaction between adjacent cells into a set of essentially independent physical effects, each of which can be modelled by an appropriate numerical method capable of modelling the structure of the corresponding wave. In this paper the emphasis has been on explicit schemes, but the value of the concept seems to be proving itself in implicit methods also [6]. A successful treatment of source terms and moving grids is described in [7], and of highly irregular grids in [2, 5, 50, 55]. Another possible application is to somewhat unorthodox explicit large time-step methods [56, 57], although I am not sure whether good numerical results for systems have yet been obtained by these means. The more modest claim that I would now propose is that systems of equations can be effectively solved, provided good numerical methods are available for scalar equations which exhibit the same modes of nonlinear behaviour.

Application of these ideas to multidimensional problems has been largely through the operator splitting concept [2, 3, 4, 6, 11, 40, 51]. The widespread feeling that it should be possible to improve on this is only slowly finding expression in algorithms. What has been done [2, 28, 35, 41]

falls some distance short of the target, but the difficulty lies perhaps in the great variety of possibilities, and the need to find systematic methods of exploring them.

Perhaps the most valuable outcome of making a really clean job of the one-dimensional methods would be if it turned out to provide a good springboard into higher dimensions. Meanwhile the one-dimensional methods continue to evolve out of experience to a greater extent than they are deduced from axioms. I find the following quotation very apt [**58**]: "Science builds a house with bricks, which, once laid, are not touched again. Philosophy tidies a room, and so has to handle things many times. The essence of its procedure is that it starts with a mess, but we do not mind being hazy so long as the haze gradually clears".

References

1. P. L. Roe, *The use of the Riemann problem in finite difference schemes*, Proc. Seventh Internat. Conf. on Numerical Methods in Fluid Dynamics, Stanford, 1980 (W. C. Reynolds and R. W. MacCormack, eds.), Lecture Notes in Phys., No. 141, Springer-Verlag, Berlin and New York, 1981, 354–359.

2. _____, *Fluctuations and signals, a framework for numerical evolution problems*, Numerical Methods for Fluid Dynamics (K. W. Morton and M. J. Baines, eds.), Academic Press, New York, 1982, 219–257.

3. _____, *Approximate Riemann solvers, parameter vectors, and difference schemes*, J. Comput. Phys. **43**, (1981), 357–372.

4. A. Harten, *High resolution schemes for hyperbolic conservation laws*, J. Comput. Phys. **49** (1983), 357–393.

5. A. Harten and J. M. Hyman, *A self-adjusting grid for the computation of weak solutions of hyperbolic conservation laws*, J. Comput. Phys. **50** (1983), 235.

6. H. C. Yee, R. F. Warming and A. Harten, *Implicit total variation diminishing (TVD) schemes for steady state calculations*, NASA TM 84342, March 1983.

7. M. Pandolfi, *A contribution to the numerical prediction of unsteady flows*, AIAA Paper 83-0121, January 1983.

8. C. K. Lombard, *Conservative supra-characteristics method for splitting hyperbolic systems of gasdynamics for real and perfect gases*, NASA CR 166307, January 1982.

9. C. K. Lombard, J. Oliger and J. Y. Yang, *A natural conservative flux-difference splitting for the hyperbolic systems of gasdynamics*, AIAA Paper 82-0976, 1982.

10. P. Colella, *Approximate solution of the Riemann problem for real gases*, preprint, Lawrence Berkeley Laboratory, University of California, May 1982.

11. S. Osher and F. Soloman, *Upwind schemes for hyperbolic systems of conservation laws*, Math. Comp. **38** (1982), 339–374.

12. S. Osher, *Shock modelling in aeronautics*, Numerical Methods for Fluid Dynamics (K. W. Morton and M. J. Baines, eds.), Academic Press, New York, 1982, 179–217.

13. P. L. Roe and J. Pike, *Efficient construction and utilisation of approximate Riemann solvers* (Comput. Methods Appl. Science Engrg., VI, R. Glowinski and J. L. Lions, eds.), North-Holland, 1984, 518.

14. B. van Leer, *On the relation between the upwind-differencing schemes of Godunov, Engquist-Osher and Roe*, SIAM J. Sci. Statist. Comput. **5** (1984), 1–20.

15. S. K. Godunov, *A difference method for the numerical calculation of discontinuous solutions of hydrodynamic equations*, Mat.-Sb. **47** (1959); English transl. U. S. Department of Commerce, JPRS 7225, 1960.

16. P. L. Roe, *On the internal structure of some captured shockwaves*, RAE report, in preparation.

17. R. F. Warming and R. W. Beam, *Upwind second-order difference schemes and applications in aerodynamic flows*, AIAA July 14, 1976.

18. T. H. Pulliam and J. L. Steger, *On implicit finite-difference simulations of three-dimensional flows*, AIAA Paper 78–10, 1978.

19. A. Jameson, W. Schmidt and E. Turkel, *Numerical solution of the Euler equations by finite-volume methods using Runge–Kutta time-stepping schemes*, AIAA Paper 81–1259, 1981.

20. A. Jameson, *Transonic aerofoil calculations using the Euler equations*, Numerical Methods in Aeronautical Fluid Dynamics (P. L. Roe, ed.), Academic Press, New York, 1982, 289–308.

21. B. van Leer, *Towards the ultimate conservative differencing scheme*. Part I, Proc. 3rd Intl. Conf. Num. Meth. in Fluid Dynamics.

22. _____, *Towards the ultimate conservative differencing scheme*. Part II, J. Comput. Phys. **14** (1974), 361–370.

23. _____, *Towards the ultimate conservative differencing scheme*. Part III, J. Comput. Phys. **23** (1977), 263–275.

24. _____, *Towards the ultimate conservative differencing scheme*. Part IV, J. Comput. Phys. **23** (1977), 276–299.

25. _____, *Towards the ultimate conservative differencing scheme*. Part V, Comput. Phys. **32** (1979), 101–136.

26. P. L. Roe, *An improved version of MacCormack's shock-capturing algorithm*, RAE TR 79041, 1979.

27. _____, *Numerical algorithms for the linear wave equation*, RAE TR 81047, 1981.

28. P. L. Roe and M. J. Baines, *Algorithms for advection and shock problems*, Proc. 4th GAMM Conf. on Numerical Methods in Fluid Mechanics, Paris, 1981 (H. Viviand, ed.), Vieweg, Braunschweig, 1982, 281–290.

29. B. Gabutti, *On two upwind finite-difference schemes for hyperbolic equations in non-conservative form*, Comput. & Fluids **11** (1983), 207–230.

30. P. Colella, *A direct Eulerian MUSCL scheme for gasdynamics*, SIAM J. Sci. Statist. Comput. (to appear).

31. P. Colella and P. R. Woodward, *The piecewise parabolic method (PPM) for gas dynamical simulations*, J. Comput. Phys. **54** (1984), 174–201.

32. M. Abramowitz and Irene Stegun, *Handbook of mathematical functions*, U. S. Department of Commerce, Washington, D. C., 1964.

33. G. D. van Albada, B. van Leer and W. W. Roberts, *A comparative study of computational methods in cosmic gas dynamics*, ICASE Report 81–24, August 1981.

34. W. A. Mulder and B. van Leer, *Implicit upwind methods for the Euler equations*, AIAA paper 83–1930, 1983.

35. P. L. Roe and M. J. Baines, *Asymptotic behaviour of some non-linear schemes for linear advection*, Proc. 5th GAMM Conf. Num. Meth. in Fluid Mechanics, Rome, 1983 (M. Pandolfi and R. Piva, eds.), Vieweg, 1984, 283–290.

36. P. K. Sweby, *High resolution schemes using flux limiters for hyperbolic conservation laws*, preprint, University of California, Los Angeles, 1983.

37. G. A. Sod, *A survey of several finite difference methods for systems of non-linear hyperbolic conservation laws*, J. Comput. Phys. **27** (1978), 1.

38. S. Chakravarthy and S. Osher, *Computing with high-resolution upwind schemes for hyperbolic equations*, Large Scale Computations in Fluid Mechanics (Proc. 15th AMS-SIAM Summer Seminar on Applied Mathematics, La Jolla, 1983, B. J. Engquist, S. J. Osher and R. C. J. Somerville, eds.), AMS, Providence, 1985.

39. P. Woodward and P. Colella, *The numerical simulation of two-dimensional fluid flow with strong shocks*, J. Comput. Phys. **54** (1984), 115–173.

40. C. C. L. Sells, *Solution of the Euler equations for transonic flow past a lifting aerofoil*, RAE TR 80065, 1980.

41. C. C. Lytton, *Solution of the Euler equations for transonic flow past a tilting airfoil—the Bernouilli formulation*, RAE TR 84080, 1984.

42. R. C. Lock, *Test cases for numerical methods in two-dimensional transonic flows*, AGARD Report No. 575, 1970.

43. M. D. Salas, *Recent developments in transonic Euler flow over a circular cylinder*, NASA TM 83282, April 1982.

44. G. K. Batchelor, *On steady laminar flow with closed streamlines at large Reynolds number*, J. Fluid Mech. **1** (1956), 177–190.

45. A. Harten, J. M. Hyman and P. D. Lax, *On finite difference approximation and entropy conditions for shocks*, Comm. Pure Appl. Math. **29** (1976), 297–322.

46. A. Harten, P. D. Lax and B. van Leer, *On upstream differencing and Godunov-type schemes for hyperbolic conservation laws*, SIAM Rev. **25** (1983), 35–60.

47. P. D. Lax, *Weak solutions of non-linear hyperbolic equations and their numerical computation*, Comm. Pure Appl. Math. **7** (1954), 159–193.

48. S. Osher, *Riemann solvers, the entropy condition, and difference approximations*, SIAM J. Numer. Anal. **21** (1984), 217–235.

49. A. Harten and P. D. Lax, *A random-choice finite-difference scheme for hyperbolic conservation laws*, SIAM J. Numer. Anal. **18** (1981), 289–315.

50. P. K. Sweby, *Shock-capturing schemes*, Ph. D. Thesis, University of Reading, Reading, U. K., 1982.

51. P. Glaister, Private communication, University of Reading, 1983.

52. H. Polachek, *Shock wave interactions*, High Speed Aerodynamics and Jet-Propulsion. III. Fundamentals of Gas Dynamics, Section E (H. W. Emmons, ed.), Oxford Univ. Press, New York and London, 1958, 482–522.

53. L. F. East, *The application of laser anemometry to the investigation of shockwave-boundary layer interactions*, AGARD CPP 193, alos RAE TM 1666, 1976.

54. W. D. Hayes and R. F. Probstein, *Hypersonic flow theory*, Vol. I. Inviscid Flows, Academic Press, New York, 1966.

55. J. Pike, *Euler equation solutions on irregular grids*, RAE report, in preparation.

56. R. J. LeVeque, *A large time step generalisation of Godunov's method for systems of conservation laws*, preprint, University of California, Los Angeles, 1984.

57. A. Harten, *On a large time-step high resolution scheme*, ICASE Report 82–34, November 1982.

58. L. Wittgenstein, *Lectures, Cambridge* 1930–32 (D. Lee, ed.), Blackwell, Oxford, 1980.

ROYAL AIRCRAFT ESTABLISHMENT, BEDFORD, UNITED KINGDOM

Current address: College of Aeronautics, Cranfield Institute of Technology, Cranfield Bedford MK43 OAL, England